

MT1003 Calculus and Analytical Geometry

Tuesday, April 11, 2023

Course Instructor

Mr. Muhammad Munawar

Mr. Muhammad Shoaib

Mr. Sameer Munir

Solution

Serial No:

2nd Sessional Exam

Total Time: 60 Minutes

Total Marks: 40

Signature of Invigilator

Roll No

Section

Signature

DO NOT OPEN THE QUESTION BOOK OR START UNTIL INSTRUCTED.

Instructions:

1. Verify at the start of the exam that you have a total of four (04) questions printed on Nine (09) pages including this title page.
2. Attempt all questions on the question-book and in the given order.
3. The exam is closed books, closed notes. Please see that the area in your threshold is free of any material classified as 'useful in the paper' or else there may a charge of cheating.
4. Read the questions carefully for clarity of context and understanding of meaning and make assumptions wherever required, for neither the invigilator will address your queries, nor the teacher/examiner will come to the examination hall for any assistance.
5. Fit in all your answers in the provided space. You may use extra space on the backside of the pages if required. If you do so, clearly mark question/part number on that page to avoid confusion.
6. Use only your own stationery and calculator. If you do not have your own calculator, use manual calculations.
7. Use only permanent ink-pens. Only the questions attempted with permanent ink-pens will be considered. Any part of paper done in lead pencil cannot be claimed for checking/rechecking.
8. Extra sheet is not allowed.

	Q-1	Q-2	Q-3	Q-4	Total
Total Marks	10	10	10	10	40
Marks Obtained					

Vetted By: _____ Vetter Signature: _____

University Answer Sheet Required: No ☐ Yes ☐

The force F (in Pound) is acting at an angle θ with the horizontal that is need to drag a crate of weight W along a horizontal surface at a constant velocity is given by

$$F = \frac{\mu W}{\cos \theta + \mu \sin \theta}$$

Where μ is a constant called the coefficient of sliding friction between the crate and the surface. Suppose that the crate weighs 150 lb. and that $\mu = 0.3$

- Find $\frac{dF}{d\theta}$ when $\theta = 30$. Express the answer in units of pounds/degree.
- Find $\frac{dF}{dt}$, when θ is decreasing at the rate of $0.5^\circ/\text{s}$ at this instant
- Let $0 \leq \theta \leq \frac{\pi}{2}$ Show that F is minimized when $\mu = \tan \theta$

Q. Solution:

$$\begin{aligned} \frac{dF}{d\theta} &= \frac{d}{d\theta} (\cos \theta + \mu \sin \theta)^{-1} \\ &= -\mu W (\cos \theta + \mu \sin \theta)^{-2} (-\sin \theta + \mu \cos \theta) \\ \frac{dF}{d\theta} &= \frac{-\mu W (-\sin \theta + \mu \cos \theta)}{(\cos \theta + \mu \sin \theta)^2} \\ \frac{dF}{d\theta} &= \frac{-(0.3)(150)(-\sin 0^\circ + 0.3 \cos 0^\circ)}{\cos^2 0^\circ + 0.3 \sin^2 0^\circ} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \frac{dF}{dt} &= \mu W \frac{d}{dt} (\cos \theta + \mu \sin \theta)^{-1} \\ &= -\mu W (\cos \theta + \mu \sin \theta)^{-2} \left(-\sin \theta \frac{d\theta}{dt} + \mu \cos \theta \frac{d\theta}{dt} \right) \\ &= \frac{-\mu W (-\sin \theta + \mu \cos \theta) \frac{d\theta}{dt}}{(\cos \theta + \mu \sin \theta)^2} \\ \frac{d\theta}{dt} &= -0.5 \end{aligned}$$

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$$\frac{dF}{dt} = \frac{-0.3 \times 150 (-\sin 30^\circ + 0.3 \cos 30^\circ)}{(\cos 30^\circ + \mu \sin 30^\circ)} \times (-0.5)$$

(iii) $\frac{dF}{dt} = 0$

$$\frac{-\mu W (-\sin \theta + \mu \cos \theta)}{\cos \theta + \mu \sin \theta} = 0$$

$$-\sin \theta + \mu \cos \theta = 0 \times \frac{(\cos \theta + \mu \sin \theta)}{\mu W}$$

$$\mu \cos \theta = \sin \theta$$

$$\mu = \tan \theta$$

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- a) Find the equation of tangent line to the curve $y = \frac{\sqrt{x}}{(1+x^2)}$ at the point $(1, \frac{1}{2})$.

Find slope

$$\frac{dy}{dx} = \frac{(1+x^2) \frac{1}{2\sqrt{x}} - \sqrt{x}(1+2x)}{(1+x^2)^2}$$

$$m = \left. \frac{dy}{dx} \right|_{(1, \frac{1}{2})} = \frac{(1+1) \frac{1}{2\sqrt{1}} - \sqrt{1}(1+2)}{(1+1)^2}$$

$$= \frac{1-3}{4} = -\frac{1}{2}$$

equation of line

$$y - y_1 = m(x - x_1)$$

$$(y - \frac{1}{2}) = -\frac{1}{2}(x - 1)$$

b) Evaluate $\lim_{x \rightarrow 1} \frac{x^x - x}{x - 1 - \ln x}$ •

$$\lim_{x \rightarrow 1} \frac{1-1}{1-1-0} = \left(\frac{0}{0}\right) \text{ Indeterminate form}$$

Apply L'Hopital Rule

$$\lim_{x \rightarrow 1} \frac{x^x (1 + \ln x) - 1}{1 - 0 - \frac{1}{x}}$$

$$\frac{1'(1 + \ln 0) - 1}{\left(\frac{0}{0}\right)}$$

again apply L'Hopital Rule

$$\lim_{x \rightarrow 1} \frac{x^x (1 + \ln x)^2 + \left(0 + \frac{1}{x}\right)x^x - 0}{0 + \frac{1}{x^2}}$$

apply

limit

$$\frac{1 + (1 + \ln 0)^2 + (0 + 1)1}{1}$$

$$\frac{1}{1}$$

$$\frac{1 + 1}{1} = 2$$

$$\frac{d}{dx} x^x = x^x (1 + \ln x)$$

$$x x^{x-1}$$

For what values of x ,

- a) Find the point x for which the given function is not continuous. Determine whether such point is removable discontinuity, if yes find the function which make it continuous.

$$f(x) = \begin{cases} 2x - 3 & \text{if } x \leq 2 \\ x^2 & \text{if } x > 2 \end{cases}$$

Identify the type of discontinuity and if possible find ~~the~~ continuous extension of $f(x)$ where

$$f(x) =$$

LHL

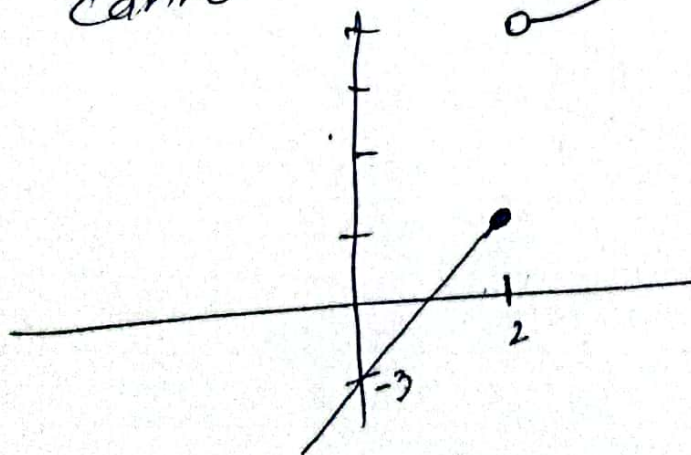
$$\lim_{x \rightarrow 2^-} 2x - 3 = 4 - 3 = 1$$

RHL

$$\lim_{x \rightarrow 2^+} x^2 = 4$$

$$LHL = RHL$$

So there will be jump discontinuity.
it cannot be removed.



b) Find the value of a and b that make the function $f(x)$ continuous in its domain.

$$f(x) = \begin{cases} \frac{x^2 - 4}{x - 2} & x < 2 \\ ax^2 + bx + 3 & 2 \leq x \leq 3 \\ 2x - a - b & x > 3 \end{cases}$$

LHL given function is continuous at $x=2$ and $x=3$

$$\lim_{x \rightarrow 2^-} \frac{x^2 - 4}{x - 2} = 2$$

RHL

$$\lim_{x \rightarrow 2^+} 4a + 2b + 3$$

$$LHL = RHL$$

$$4a + 2b + 3 = 2$$

$$4a + 2b + 1 = 0 \quad \text{--- (1)}$$

$$\lim_{x \rightarrow 3^-} 9a + 3b + 3$$

$$LHL \lim_{x \rightarrow 3^-} ax^2 + bx + 3 = 9a + 3b + 3$$

$$RHL \lim_{x \rightarrow 3^+} 2x - a - b = 6 - a - b$$

$$RHL = LHL$$

$$9a + 3b + 3 = 6 - a - b$$

$$10a + 4b - 3 = 0 \quad \text{--- (2)}$$

$$2 \times (1) = (2)$$

$$8a + 4b + 2 = 0$$

$$10a + 4b - 3 = 0$$

$$\underline{\hspace{10em}}$$

$$-2a - 1 = 0$$

$$\boxed{a = -\frac{1}{2}}$$

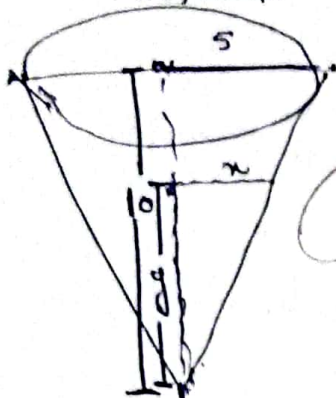
$$(1) \Rightarrow 2 \times (-\frac{1}{2}) + 2b + 1 = 0$$

$$-2 + 2b + 1 = 0$$

$$2b = 1$$

$$\boxed{b = \frac{1}{2}}$$

- a) Water runs into a conical tank at the rate of $9 \text{ ft}^3/\text{min}$. The tank stands point down and has a height of 10 ft and a base of radius 5 ft . How fast is the water level rising when the water is 6 ft deep?



given

$$\frac{dv}{dt} = 9 \text{ ft}^3/\text{min} ; \frac{dy}{dt} = ?$$

$$H = 10$$

$$R = 5$$

by using similar Δ

$$\frac{x}{y} = \frac{R}{H}$$

$$x = y/2$$

$$V = \frac{1}{3} \pi x^2 y$$

$$= \frac{1}{3} \pi (y/2)^2 y$$

$$V = \frac{1}{12} \pi y^3$$

$$\frac{dv}{dt} = \frac{\pi}{12} y^2 \frac{dy}{dt}$$

$$\frac{dv}{dt} = \frac{\pi}{12} (y)^2 \frac{dy}{dt}$$

$$9 = \frac{\pi}{12} (16)^2 \frac{dy}{dt}$$

$$9 = \frac{\pi}{12} (256) \frac{dy}{dt}$$

$$\frac{1}{\pi} = \frac{dy}{dt}$$

- b) Find the horizontal and vertical asymptotes of $f(x) = -\frac{8x}{x^2-4}$ and plot the graph.

6) plot the graph of $f(x) = -\frac{8x}{x^2-4}$ and identify its Horizontal & vertical Asymptotes
For vertical asymptote

$$x^2 - 4 = 0$$

$$x = \pm 2$$

for Horizontal asymptote

$$y = \lim_{x \rightarrow \infty} \frac{-8x}{x^2-4} = \frac{-8/x^2}{x^2/x^2 - 4/x^2} = 0$$

